

Practical Exam – Outbound Nursing Team

From:	Manager, Analytics
Received:	Today
Subject:	New task from the Outbound Call Nursing team
Hey! I have an analysis task for you from the Outbound Call Nursing team. I think you will find it interesting, there is lots of flexibility for you to try different methods. You can see the request in the email below. I would like you to perform the analysis and write a short report for me. I don't need to see any code, but I do want to read what you were thinking about as you did the analysis and how you got to your conclusions. I also want you to prepare and deliver the presentation to the Manager of the Outbound Call Nursing Team-you are ready for the challenge! You can find more details about what I expect you to do here. And information on the data here. I will be on vacation for the next couple of weeks, but I know you can do this without my support. If you need to make any decisions, include them in your work and I will review them when I am back. Good Luck!	

From:	Manager, Outbound Call Nursing Team
To:	Manager, Analytics
Received:	Yesterday
Subject:	Effectiveness of current outbound phone call prioritization
Hi, Really hoping you can help me out with some analysis. This is quite short notice but I have a meeting with the executive team to update them on our progress, so I need this information as soon as possible. As you know, one of our key goals is to increase screening compliance rates (the number of people attending their screening appointments). We need to know: - How many patients were reached successfully? - Were there differences in compliance depending on how many screenings a patient was eligible for? - Were patients more likely to get their screenings done (i.e. become compliant) after we reach them compared to patients we do not reach? - Based on the data, how should we optimize our outbound calls to maximise compliance? We don't really know if there are other differences in approach or the patient groups, so anything you can tell us would be helpful to give some information on how to approach outreach calls effectively. I need to report to the executive team in 4 weeks. You need to present to me before then so I understand what is going on and what we do next. Look forward to seeing your presentation.	

About Universal Healthy Humans Company

As a partially government funded healthcare company, Universal Healthy Humans Company (UHHHC) is accountable for the screening compliance rate of its customers. To support customers in completing required annual healthcare screenings, UHHHC operates an outbound call center staffed by nurses who are tasked with making contact with customers who have not completed their required screenings and supporting them in understanding the importance of and gaining access to resources to complete their screenings.

Preventive Screening Outreach via Phone

Our clinic runs a targeted outreach initiative to boost participation in preventive health screenings for patients over the age of 50. While digital and print communications have their place, we focused exclusively on **personal phone calls** to increase patient engagement and trust.

The initiative prioritized five critical screening areas: bowel cancer (BCS), colorectal cancer (COL), controlling high blood pressure (CBP), osteoporosis management in women (OMW), and early elective delivery prevention (EED).

A member of the clinical staff called each patient to:

- Explain why the screening was recommended
- Address any concerns or misconceptions
- Help schedule necessary appointments or follow-ups

Each call lasted approximately 15 to 20 minutes, allowing time for a two-way conversation and ensuring the patient felt informed and supported.

Data Information

The Outbound Call Nurse team has provided some data about their phone call activity, patient screening eligibility and compliance activity.

You can find the data [here](#). I will let you decide how to process it, just make sure you include all your decisions in your report.

The data hasn't been validated, so make sure that you check it against all of the information in the table before you start your analysis.

Column Name	Details
patient_id	Numeric, length of 22, unique identifier for each patient
screening_type	Character, length of 3, representing the abbreviation for each healthcare screening. One of ["BCS", "COL", "EED", "CBP", "OMW"]
screening_completed_ind	Boolean, an indicator for if a patient has completed the screening [0 = not completed, 1 = completed, null = not eligible]
screening_date	Date, the date of the screening completed in the format "YYYY-MM-DD"
latest_call_date	Date, the date of the latest Outbound Call Nurse team call attempt in the format "YYYY-MM-DD"
reached_ind	Boolean, an indicator for if a patient was reached during the call [0 = not reached, 1 = reached, null = not called]

Guide to Analysis Projects

1. I would like you to create a written report to summarize the analysis you have performed and your findings. The report will be read by me (Manager, Analysis). The list below describes what I expect to see in your written report.

2. You will need to use a DataLab workbook to write up your findings and share visualizations.
3. You must use the data provided for the analysis.
4. You will also need to prepare and deliver a presentation. You should prepare around 8-10 slides to present to the Manager of the Outbound Nursing Team. The list below describes what they expect to see in your presentation.
5. Your presentation should be no longer than 10 minutes.

Written Report

Your written report should include written text summaries and graphics of the following:

- Data validation:
 - Describe validation and cleaning steps for every column in the data
- Exploratory Analysis to answer the customer questions ensuring you include:
 - Two different types of graphic showing single variables only
 - At least one graphic showing two or more variables
 - Description of your findings
- Definition of a metric for the business to monitor
 - How should the business monitor what they want to achieve?
 - Estimate the initial value(s) for the metric based on the current data
- Final summary including recommendations that the business should undertake

Presentation

You will give an overview presentation to the Manager of the Outbound Nursing Team. The presentation should include:

- An overview of the project and business goals
- A summary of the work you undertook and how this addresses the problem
- Your key findings including the metric to monitor and current estimation
- Your recommendations to the business

Grading

Before submitting your written report or delivering your presentation, remember to check your work against the grading criteria.

You can find the full grading rubric [here](#).

You must pass all criteria to pass this part of the certification.

Sample Solution

You can find a sample solution from a published workbook and a recording. The sample solution demonstrates one possible approach; others are possible.